(Please write your Enrollment Number) Enrollment No. 025

End-Term Examination

(CBCS)(SUBJECTIVE TYPE)(OffLine)

Course Name:<B.Tech IT/ CSE/ CSE-AI/ AI&ML >, Semester:<03>

(November-December, 2023)

(REGULAR& RE-APPEAR)

Subject Code: BIT 203	**Subject: Software Engineering**
Time :3 Hours	**Maximum Marks :60**

Note: Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1	Attempt any 4 parts	**(4*5=20)**	
(a)	What are software Configuration Items?		
(b)	What is Software Reverse Engineering		
(c)	What are the major uses of requirements specification		
(d)	What is the process of risk management?		
(e)	*"Well-Structured Design improve the maintainability of a System"* Comment		
	UNIT-I		
Q2	Suppose you are involved in a large project concerning the development of a patient planning system for a hospital. You may opt for one of the following two strategies. **The first strategy** is to start with a thorough analysis of user requirements, after which the system is build according to these requirements. **The second strategy** starts with a less complete requirements analysis phase, after which a pilot version is developed. The pilot version is installed in a few small departments. Further development of the system is guided by the experience gained in working with the pilot version. Discuss the pros and cons of both strategies. Which strategy do you favour?		10
Q3	Suppose you are managing a project which is getting behind schedule. Possible actions include renegotiating the time schedule, adding people to the project and renegotiating quality requirements. In which ways can these actions shorten the time schedule? Can you think of other ways to finish the project on time?		10
	UNIT-II		
Q4	**Describe the main reasons why documentation of requirements is important for many businesses?**		10
Q5	Draw the Level 1 DFD for the following "Order System" i). A customer submits an Order. Depending on the processing logic, the fill order process either sends an order reject notice back to the customer or sends a picking list to the warehouse. ii). A completed order from the warehouse is input to the create invoice process, which outputs an invoice to both the customer process and the accounts receivable data store. iii). A customer makes a payment that is processed by apply payment. Apply payment requires invoice detail input from the accounts receivable data store along with the payment. Apply payment also outputs payment detail back to the accounts receivable data store and outputs commission to the sales dept, bank deposit to the bank and cash receipts entry to accounting.		10
	UNIT-III		
Q6	**Draw the CFG for the following snippet of the program and calculate the Cyclomatic Complexity of the programme.** **begin** no_6=true; Sum=0;		10

```
for (i=1; i<no_of_courses; i++)
{
  if (grade[i]<7)
     no_6=false;
     sum= sum+grade[i];
};
Average=sum/no_of_courses;
If ((average>= 8 && (no_6))
{
System.out.println("with Distinction");
}
end;
```

Q7 **Describe the software maintenance process** 10

UNIT-IV

Q8 **Suppose you have been hired as a Project Manager for the development of University Automation System. After detailed discussions with all the stakeholders, you have identified the following complexities of the System:** 10

Components	Number of Components		
	Low Complexity	Average Complexity	High Complexity
External Inputs	12	07	05
External Output	20	25	40
External Inquiries	10	15	25
Internal Logical files	02	05	07
External Interface Files	25	15	12

In addition to above, the Project Manager has identified the following additional properties that the system must fulfil:

There will be Significant data communication. There is no influence of distributed data processing. Performance of the application is very critical. Computer Hardware is moderately utilized. The application Designed code may be moderately reusable. End User efficiency is average. System is not designed for multiple installations in different Universities. The portability of the application is critical. There will be significant transaction rate and Online updating of the data is significant. Other complexity adjustment factors are treated as average.

You being the Project Manager of the Development Team, is required to Compute the function points for the project so that the size, cost and efforts for developing the software application may be estimated

Q9 A project size of 20000 LOC is to be developed. Software development team has average experience on similar type of projects. There is medium innovation required for development of the Project. The project schedule is not very tight. Delivery deadline of the project is not very tight. 10

Calculate the effort, development time, average staff size and productivity of the project.

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